IN THE NON-BIPARTITE CASE THE SET A OF THE GALLAI-EDMONDS DECOMPOSITION STAYS BELOW THE THRESHOLD THROUGHOUT THE WINDOW

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned author source sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the source pins the publisher class scrartcl, which the delivery wrapper does not accept, so this card uses sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/retained-preamble.tex, which supplies the amsart baseline and then carries the author's own newtheorem and macro declarations (pinned source lines 30-57) verbatim. No macro was invented and no mathematical symbol was changed.

Editorial note. Bibliography provenance: the source carries no thebibliography environment and ends with bibliographystyle abbrv plus bibliography perc (author source lines 1691-1692), so the bibliography is resolved with BibTeX from the identical copy staged with the same arXiv version under sources/product-graph-matchings-2404.14020v4/perc.bib; no entry was edited and the bibliography style is the author's own abbrv.

Editorial note. The counted proof is the longest in the paper and relies on the author's lemmas on the exact boundary identity (source lines 1099-1103), on the strict Hall inequality (source lines 1073-1079) and on the weighted counting bound for bipartitions (source lines 1344-1364); all three have their own cards in this lane, so here they are navigation pointers only and their content is declared as named standard prerequisites.

Editorial note. The threshold u_0 that the statement compares against is delivered verbatim as a context span, quoted from the author's own display.

Allowed standard prerequisites (editorial).

- The Gallai-Edmonds decomposition (D, A, C), its parameters m, a, r and h, and the auxiliary bipartite graph Q between A and the components of the D part. Set up in the author's subsection on the decomposition (source lines 1054-1098). Fixes every object the proof manipulates during the counting.
- The exact edge-counting identity for the Gallai-Edmonds parameters. The author's lemma at source lines 1099-1103, whose proof is delivered by this lane's separate card for that statement. Turns the assumed size of A into an identity relating the boundary of D, the internal edges of A and the component sizes.
- The strict Hall inequality for the auxiliary bipartite graph, and the weighted counting bound for bipartitions with a small boundary. The author's lemmas at source lines 1073-1079 and 1344-1364, both delivered by this lane's separate cards. Control respectively the matching structure around A and the number of bipartitions of the relevant set, which together give the probabilistic contradiction.

Source reference. Original lemma 7.4: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 1099–1099. Navigation pointer only: the counted proof uses this exact boundary identity in its main estimate; the identity's own statement and proof are delivered by this lane's separate card.

Source reference. Original lemma 7.2: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 1073–1073. Navigation pointer only: the counted proof uses this strict Hall inequality for the auxiliary bipartite graph; its own card exists in this lane and its text is not reproduced here.

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Source reference. Original lemma 7.7: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 1344–1344. Navigation pointer only: the counted proof uses this weighted counting bound for bipartitions in its final step; its own card exists in this lane.

Source reference. Original equation 7.3: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 1360–1360. Navigation pointer only: it labels the displayed weighted consequence of the counting lemma that the counted proof applies at the end; the display belongs to that other statement and is not reproduced here.

Source reference. Original equation 7.4: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 1100–1100. Navigation pointer only: the counted proof consumes the estimate or statement labelled here; its content is declared as a named standard prerequisite and the labelled text is not reproduced in this card.

Source reference. Original equation 7.5: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 814–814. Navigation pointer only: the counted proof consumes the estimate or statement labelled here; its content is declared as a named standard prerequisite and the labelled text is not reproduced in this card.

Source reference. Original equation 1.1: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 224–224. Navigation pointer only: the counted proof consumes the estimate or statement labelled here; its content is declared as a named standard prerequisite and the labelled text is not reproduced in this card.

Source reference. Original lemma 2.2: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 318–318. Navigation pointer only: the counted proof consumes the estimate or statement labelled here; its content is declared as a named standard prerequisite and the labelled text is not reproduced in this card.

Source reference. Original lemma 2.3: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 231–231. Navigation pointer only: the counted proof consumes the estimate or statement labelled here; its content is declared as a named standard prerequisite and the labelled text is not reproduced in this card.

Source reference. Original lemma 6.7: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 811–811. Navigation pointer only: the counted proof consumes the statement or estimate labelled here; its content is declared as a named standard prerequisite and the labelled text is not reproduced in this card.

Source reference. Original lemma 6.3: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 714–714. Navigation pointer only: the counted proof consumes the statement or estimate labelled here; its content is declared as a named standard prerequisite and the labelled text is not reproduced in this card.

Source reference. Original lemma 6.2: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 703–703. Navigation pointer only: the counted proof consumes the statement or estimate labelled here; its content is declared as a named standard prerequisite and the labelled text is not reproduced in this card.

Source reference. Original equation 6.1: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 660–660. Navigation pointer only: the counted proof consumes the statement or estimate labelled here; its content is declared as a named standard prerequisite and the labelled text is not reproduced in this card.

Source reference. Original theorem 2.1: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 220–220. Navigation pointer only: the counted text refers to the statement or display labelled here; its content is declared as a named standard prerequisite and the labelled text is not reproduced in this card.

Source reference. Original equation 7.2: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 1378–1378. Navigation pointer only: the counted text refers to the statement or display labelled here; its content is declared as a named standard prerequisite and the labelled text is not reproduced in this card.

Lemma 0.1. *Suppose that at least one base graph of G is non-bipartite. Whp for every $p \in [p_-, p_+]$, if $\operatorname{def}(G_p) \geq 2$, then the set A in the Gallai–Edmonds decomposition of G_p has size at most u_0.*

Proof. Fix $p \in [p_-, p_+]$, let (D, A, C) be the Gallai–Edmonds decomposition of G_p, and suppose that $\operatorname{def}(G_p) \geq 2$ and $|A| > u_0$. Let D_0 be the union of the singleton components of $G_p[D]$, let D^+ be the family of non-singleton components of $G_p[D]$, and put

$$D_1 = \bigcup_{D_i \in D^+} D_i.$$

Write $a = |A|$, $r = |D^+| + |D_0| - a \geq 2$, and recall

$$h = \sum_{i<j} e_G(D_i, D_j) + e_G(D, C).$$

All the edges counted in h are not present in G_{p_-}. Hence, we shall sum $(1-p_-)^h$ over the choices of A, C, the singleton components of D, and the non-singleton components of D.

Every $D_i \in D^+$ is a non-singleton factor-critical component, so $|D_i| \geq 3$. By (7.4),

$$h = \frac{dr}{2} + e_G(A) + \frac{e_G(C, C^c)}{2} + \frac{1}{2} \sum_{D_i \in D^+} \big(e_G(D_i, D_i^c) - d\big). \tag{1}$$

Also

$$h \geq e_G(D_0), \tag{2}$$

since every edge of $G[D_0]$ joins two distinct components of $G_p[D]$. Taking three quarters of (1) and one quarter of (2) gives

$$h \geq \frac{3dr}{8} + \frac{3}{4} e_G(A) + \frac{1}{4} e_G(D_0) + \frac{3}{8} e_G(C, C^c) + \frac{3}{8} \sum_{D_i \in D^+} \big(e_G(D_i, D_i^c) - d\big). \tag{3}$$

Put $k_i = \min\{|D_i|, |D_i^c|\}$ so that $3 \leq k_i \leq n/2$. If $k_i \leq 2u_1$, then (7.5) gives $e_G(D_i, D_i^c) \geq 2d$. If $k_i > 2u_1$, then $\ln(n/k_i) \geq \ln 2$, and (1.1) gives

$$e_G(D_i, D_i^c) \geq \frac{e \ln 2}{M} k_i > 2d$$

since $u_1 \gg d$. Thus,

$$\frac{1}{8}\big(e_G(D_i, D_i^c) - d\big) \geq \frac{1}{16} e_G(D_i, D_i^c). \tag{4}$$

Furthermore,

$$e_G(C, C^c) + \sum_{D_i \in D^+} e_G(D_i, D_i^c) \geq e_G((C \cup D_1), (C \cup D_1)^c). \tag{5}$$

Plugging (4) and (5) into (3) we obtain the following lower bound for h:

$$h \geq \frac{3dr}{8} + \frac{e_G(A) + e_G(D_0)}{4} + \frac{e_G((C \cup D_1), (C, \cup D_1)^c)}{16} + \frac{1}{4} \sum_{D_i \in D^+} \big(e_G(D_i, D_i^c) - d\big). \tag{6}$$

We first sum over the possible families D^+ for a fixed set $(C \cup D_1)$. By Original lemma 2.3 here are at most $|C \cup D_1|(ed)^{k-1}$ connected sets of cardinality k with a root in $C \cup D_1$.

For $3 \leq |D_i| \leq 2u_1$, Original lemma 6.7 gives

$$e_G(D_i, D_i^c) - d \geq d + 15(M-1)|D_i| \ln d.$$

Plugging this into the last term of (6) the total contribution from these sets is at most

$$\begin{aligned} &|C \cup D_1| \sum_{k=3}^{2u_1} (ed)^{k-1} (1-p_-)^{\frac{d+15(M-1)k\ln d}{4}} \\ \leq &n^{3/4} \omega^{1/4} \sum_{k \geq 3} (ed)^{k-1} d^{-\frac{15\ln 2}{4} k} \\ \leq &n^{3/4+o(1)}, \end{aligned}$$

where we used that $\ln n \geq \frac{\ln 2}{M-1} d$ and that $\frac{15 \ln 2}{4} > 1$.

For $2u_1 < |D_i| \leq n/2$, first suppose that

$$e_G(D_i, D_i^c) \geq \frac{5(M-1)}{\ln 2} |D_i| \ln d.$$

Using again $\ln n \geq \frac{\ln 2}{M-1} d$, the contribution from this range is at most

$$n(1-p_-)^{-d/4} \sum_{k>2u_1} (ed)^k (1-p_-)^{\frac{5(M-1)}{4\ln 2} k \ln d}$$
$$\leq \left(\frac{\omega}{n}\right)^{-1/4} \sum_{k>2u_1} (ed)^k d^{-\frac{5}{4}k} \leq \exp\{-\Omega(u_1 \ln d)\}.$$

Indeed, $\frac{5}{4} > 1$ makes the series geometric with exponent $-\Omega(k \ln d)$, while the prefactor has logarithm $O(d)$ and $u_1 \gg d$.

Otherwise if $e_G(D_i, D_i^c) < |D_i| \ln^2 d$, then Original lemma 6.3 (which is applicable because $2u_1 \geq \frac{n}{d^{\ln^2 d}}$) gives at most $\exp\{2|D_i|/\ln d\}$ choices, whereas

$$e_G(D_i, D_i^c) - d = \Omega(|D_i|)$$

since $|D_i| > 2u_1 \gg d$. Thus,

$$\sum_{k>2u_1} \exp\left\{\frac{2k}{\ln d}\right\} (1-p_-)^{\Omega(k)}$$
$$\leq \sum_{k>2u_1} \exp\{-\Omega(k)\} = \exp\{-\Omega(u_1)\}.$$

There is at most one D_i with $|D_i| > n/2$. We now do a similar argument focusing on its complement D_i^c. Since $A \subseteq K^c$, we have $u_0 < |D_i^c| < n/2$. If

$$e_G(D_i, D_i^c) \geq \frac{8(M-1)}{\ln(2e)} |D_i^c| \ln(en/|D_i^c|),$$

then,

$$\sum_{u_0<k<n/2} \binom{n}{k} (1-p_-)^{\left\{\frac{8(M-1)}{\ln(2e)} k \ln(en/k) - d\right\}/4} = \exp(-\Omega(u_0)),$$

using that $\log n \geq \frac{\log 2}{M-1} d$ and $2 \log 2 > 1$. Otherwise if $e_G(D_i, D_i^c) \leq |D_i^c| \ln^2 d$, then by Original lemma 6.3 (which is applicable because $u_0 \geq \frac{n}{d^{\ln^2 d}}$) and $e_G(D_i, D_i^c) - d \geq \Omega(|D_i^c|)$. Here again $|D_i^c| > u_0 \gg d$ permits the subtraction of d from the linear expansion bound. Then,

$$\sum_{u_0<k<n/2} \exp\{2k/\ln d\} (1-p_-)^{\Omega(k)} = \exp(-\Omega(u_0)).$$

Let $\mathcal{S}_{\leq n/2}$ be the sum of the terms corresponding to connected sets of size at most $n/2$, and let $\mathcal{S}_{>n/2}$ be the corresponding sum for a possible connected set larger than $n/2$. The preceding estimates give

$$\mathcal{S}_{\leq n/2} \leq n^{3/4+o(1)}, \qquad \mathcal{S}_{>n/2} \leq e^{-\Omega_M(u_0)}.$$

A pairwise disjoint family contains at most one member larger than $n/2$. In particular, uniformly for sets Y with $|V \setminus Y| > u_0$, the sum over unordered families of pairwise disjoint connected subsets of $G[Y]$ satisfies

$$\sum_{\substack{D^+ \\ |V \setminus (C \cup D_1)| > u_0}} (1-p_-)^{\frac{1}{4} \sum_{D_i \in D^+} \{e_G(D_i, D_i^c) - d\}} \leq \exp\{n^{3/4+o(1)}\} = \exp\{o(u_0)\}. \tag{7}$$

This sum counts only the non-singleton components of $G_p[D]$; it does not use the vertices or internal edges of C. Given $C \cup D_1$, an ordered partition $V \setminus (C \cup D_1) = A \sqcup D_0$, and the family D^+, the remaining set $C = (C \cup D_1) \setminus \bigcup_{D_i \in D^+} D_i$ is determined.

Recalling (6), after (7) and after discarding $(1-p_-)^{3dr/8} \leq 1$, it remains to bound the following sum over ordered partitions $V = (C \cup D_1) \sqcup A \sqcup D_0$:

$$\sum_{\substack{V = (C \cup D_1) \sqcup A \sqcup D_0 \\ |A| > u_0}} (1-p_-)^{e_G((C \cup D_1), (C \cup D_1)^c)/16 + \{e_G(A) + e_G(D_0)\}/4}. \tag{8}$$

We split according to whether $s := |(C \cup D_1)^c| \leq n/2$. Suppose first that this inequality holds. Then $s \geq |A| > u_0$ and $\ln\left(\frac{en}{s}\right) = O(\ln d)$. If

$$e_G((C \cup D_1), (C \cup D_1)^c) \geq \frac{20(M-1)}{\ln 2} s \ln\left(\frac{en}{s}\right), \tag{9}$$

then

$$\binom{n}{s}2^s(1-p_-)^{e_G((C\cup D_1),(C\cup D_1)^c)/16}\le \exp\{-\Omega(s\ln(en/s))\}.$$

If (9) fails, then

$$e_G((C\cup D_1),(C\cup D_1)^c)=O(s\ln d)<s\ln^2 d.$$

By Original lemma 6.3, there are $e^{o(s)}$ choices for $(C\cup D_1)^c$, while expansion (Original theorem 2.1) gives $e_G((C\cup D_1),(C\cup D_1)^c)=\Omega(s)$. For each fixed $(C\cup D_1)^c$, Original lemma 7.7 gives

$$\sum_{A\sqcup D_0}(1-p_-)^{\{e_G(A)+e_G(D_0)\}/4}\le e^{o(s)},$$

obtained from (7.3) with $\theta=1/4$.

Thus, the range $s\le n/2$ contributes at most

$$\sum_{s>u_0}e^{-\Omega(s)}=e^{-\Omega(u_0)}. \tag{10}$$

In particular, when the Gallai–Edmonds set $C\cup D_1$ contains more than half of vertices, we count the smaller set $A\cup D_0$ rather than $C\cup D_1$.

We may now assume $s=|(C\cup D_1)^c|>n/2$, so $|C\cup D_1|<n/2$. Fix $\varepsilon>0$ to be chosen sufficiently small. Suppose first that $|C\cup D_1|\ge\varepsilon n$. If $e_G((C\cup D_1),(C\cup D_1)^c)\ge\frac{20(M-1)\ln 3}{\ln 2}n$, then

$$3^n(1-p_-)^{e_G((C\cup D_1),(C\cup D_1)^c)/16}\le e^{-\Omega(n)}.$$

Otherwise,

$$e_G((C\cup D_1),(C\cup D_1)^c)=O(n)=O(|C\cup D_1|)<|C\cup D_1|\ln^2 d.$$

By Original lemma 6.3, there are $e^{o(n)}$ choices for $(C\cup D_1)^c$, and, for each fixed $(C\cup D_1)^c$, (7.3) with $\theta=1/4$ (which is applicable since $e_G((C\cup D_1),(C\cup D_1)^c)<\frac{20(M-1)\ln 3}{\ln 2}n\le\frac{40(M-1)\ln 3}{\ln 2}|(C\cup D_1)^c|$) gives

$$\sum_{R=A\sqcup D_0}(1-p_-)^{\{e_G(A)+e_G(D_0)\}/4}\le e^{o(n)}.$$

Moreover,

$$e_G((C\cup D_1),(C\cup D_1)^c)=\Omega(|C\cup D_1|)\ge\alpha_M\varepsilon n,$$

for some suitable constant α_M. So, in total this contributes $e^{-\Omega(n)}$.

Finally suppose that $|C\cup D_1|<\varepsilon n$. Fix i such that H_i is non-bipartite. Fixing all coordinates other than the i-th partitions V into $n/|V(H_i)|$ sets, each of which induces a copy of H_i. At most $|C\cup D_1|$ of these copies meet $C\cup D_1$. In every remaining copy, the vertices lie in $A\cup D_0$; since H_i is non-bipartite, at least one edge has both endpoints in A or both endpoints in D_0. Thus

$$e_G(A)+e_G(D_0)\ge n/M-|C\cup D_1|\ge n/(2M). \tag{11}$$

The ordered partitions for which $e_G(A)+e_G(D_0)\ge\frac{5(M-1)\log 3}{\log 2}n$ contribute at most

$$3^n(1-p_-)^{\frac{5(M-1)\log 3}{4\log 2}n}=e^{-\Omega(n)}.$$

For the remaining ordered partitions with $e_G(A)+e_G(D_0)<\frac{5(M-1)\log 3}{\log 2}n$, repeat the counting argument from Original lemma 7.7 for the ordered partition $R=A\sqcup D_0$. The trivial bound gives $e_G((C\cup D_1),(C\cup D_1)^c)\le d|C\cup D_1|$, so (7.2) gives only $\varepsilon n+O(n/d)$ exceptional vertices in X. As in the proof of Original lemma 7.7, the random sample T, together with $A\triangle N_{G[R]}(T)$ after possibly interchanging A and D_0, determines the ordered partition of R. We get

$$|A\triangle N_{G[R]}(T)|\le\epsilon n+o(n),\qquad |T|=o(n).$$

The number of choices for $(T,A\triangle N_{G[R]}(T))$ is at most

$$\exp\{\varepsilon\ln\left(\frac{e}{\varepsilon}\right)n+o(n)\}$$

and the choices of $(C\cup D_1)$ contribute at most $\exp\{\varepsilon\ln(e/\varepsilon)n\}$. Recalling that by (11) there are at least $(e_G(A)+e_G(D_0))/4\ge n/(8M)$ edges that are not present in G_{p_-}. Choosing $\varepsilon=\varepsilon(M)>0$ sufficiently

small ensures $2\varepsilon \log\left(\frac{e}{\varepsilon}\right) < \frac{\log 2}{8M(M-1)}$. Then, the contribution is at most

$$\begin{aligned}
&\exp\{\varepsilon \ln\left(\frac{e}{\varepsilon}\right) n + o(n)\}(1-p_-)^{n/(8M)} \\
\leq &\exp\{2\varepsilon \ln\left(\frac{e}{\varepsilon}\right) n - \frac{\log 2}{8M(M-1)} + o(n)\} \\
= &\exp(-\Omega(n)).
\end{aligned}$$

Combining (10) with this contribution from $s > \frac{n}{2}$ proves that (8) is at most $\exp(-\Omega(u_0)) + \exp(-\Omega(n))$.

Multiplying (7) with this bound on (8) gives $o(1)$, showing that whp there is no Gallai–Edmonds decomposition of G_p with $\mathrm{def}(G_p) \geq 2$ and $|A| > u_0$. □